

EXCEPTION PRACTICE PROGRAM- 11/08/26

QUESTIONS

Q1 
Handling ArithmeticException
10 marks

Q2 
Handling
ArrayIndexOutOfBoundsException
10 marks

Q3 
Handling
NumberFormatException
10 marks

Q4 
Handling Multiple Exceptions
10 marks

4/4 answered

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 public class Exception
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         System.out.println("Enter numerator:");
8         int numerator = sc.nextInt();
9         System.out.println("Enter denominator:");
10        int denominator=sc.nextInt();
11        try
12        {
13            int result=numerator/denominator;
14            System.out.println("Result:"+result);
15        }
16        catch(ArithmeticException e)
17        {
18            System.out.println("Error:Denominator cannot be zero.")
19        }
20        finally
21        {
```

Input

10
0

Output

```
Enter numerator:
Enter denominator:
Error:Denominator cannot be zero.
```

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4/4 answered

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2 public class ArrayIndexHandler
3 {
4     public static void main(String[] args)
5     {
6         Scanner scanner = new Scanner(System.in);
7         int[] numbers = {10, 20, 30, 40, 50};
8         try
9         {
10            System.out.print("Enter an index (0-4): ");
11            int index = scanner.nextInt();
12            System.out.println("Element at index " + index + ": " + numbers[index]);
13        }
14        catch (ArrayIndexOutOfBoundsException e)
15        {
16            System.out.println("Error: The index you entered is out of range.");
17        }
18        catch (Exception e)
19        {
20            System.out.println("Error: Invalid input.");
21        }
22    }
23 }
```

Input

5

Output

Enter an index (0-4): Error: The index you entered is outside the valid range.

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10 marks

4/4 answered

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2 class Numberformat
3 {
4     public static void main(String[] args)
5     {
6         Scanner scanner = new Scanner(System.in);
7         System.out.print("Enter a string to convert to an integer:");
8         String input = scanner.nextLine();
9         try
10        {
11            int number = Integer.parseInt(input);
12            System.out.println("Conversion successful! The integer is: " + number);
13        }
14        catch (NumberFormatException e)
15        {
16            System.out.println("Error: Invalid input. The string is not a valid integer.");
17        }
18        finally
19        {
20            scanner.close();
21        }
22    }
23 }
```

Input

123

Output

```
Enter a string to convert to an
integer: Conversion successful!
The integer is: 123
```

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QUESTIONS

Q1 ✓
Handling ArithmeticException
10 marks

Q2 ✓
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Q3 ✓
Handling
NumberFormatException
10 marks

Q4 ✓
Handling Multiple Exceptions
10 marks

4/4 answered

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```
1 import java.util.Scanner;
2 public class MultipleExceptionHandling
3 {
4     public static void main(String[] args)
5     {
6         Scanner scanner = new Scanner(System.in);
7         int[] results = new int[1];
8         try
9         {
10             System.out.print("Enter first integer: ");
11             int num1 = scanner.nextInt();
12             System.out.print("Enter second integer: ");
13             int num2 = scanner.nextInt();
14             int divisionResult = num1 / num2;
15             results[0] = divisionResult;
16             System.out.println("Result stored in array: " + results[0]);
17         }
18     }
19 }
```

Input

10 0

Output

```
Enter first integer: Enter second
integer: Error: Cannot divide by
zero!
```